

1 NumPy Complete Roadmap

(Already included in previous section)

2 Pandas Complete Roadmap

(Already included in previous section)

3 Matplotlib Complete Roadmap

1 Basic Plotting

Function	Description
<code>plt.plot(x, y)</code>	line plot
<code>plt.scatter(x, y)</code>	scatter plot
<code>plt.bar(x, height)</code>	vertical bar chart
<code>plt.barh(y, width)</code>	horizontal bar chart
<code>plt.hist(data, bins)</code>	histogram
<code>plt.boxplot(data)</code>	box plot
<code>plt.pie(data, labels)</code>	pie chart

2 Customization

Function	Description
<code>plt.title('Title')</code>	plot title
<code>plt.xlabel('X-axis')</code>	x-axis label
<code>plt.ylabel('Y-axis')</code>	y-axis label
<code>plt.legend()</code>	legend display
<code>plt.xlim(a, b)</code>	set x-axis limits
<code>plt.ylim(a, b)</code>	set y-axis limits

Function	Description
<code>plt.grid(True)</code>	add grid
<code>plt.style.use('style_name')</code>	change plot style (e.g., 'ggplot', 'seaborn')

3 Subplots & Figure Management

Function	Description
<code>plt.figure(figsize=(w,h))</code>	set figure size
<code>plt.subplot(nrows, ncols, index)</code>	multiple plots in same figure
<code>plt.subplots()</code>	create multiple subplots at once

4 Saving & Displaying

Function	Description
<code>plt.show()</code>	display plot
<code>plt.savefig('filename.png')</code>	save figure

💡 Example

```
import matplotlib.pyplot as plt

x = [1,2,3,4,5]
y = [2,3,5,7,11]

plt.plot(x, y, marker='o', color='r', label='Line')
plt.title('Line Plot Example')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.grid(True)
plt.legend()
plt.show()
```